OL Organic Letters

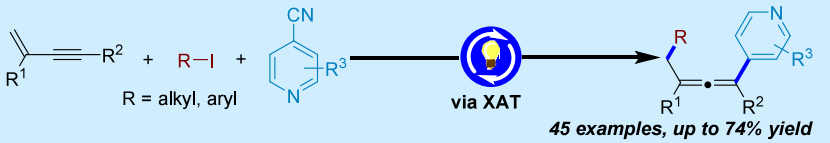
pubs.acs.org/OrgLett Letter

Photoreductive 1,4-Dicarbofunctionalization of 1,3-Enynes with Organoiodides and Cyanoarenes via Halogen-Atom Transfer

Liang-Feng Yang, Liang Zeng, Yi-Lin Liu,* Ming Hu,* and Jin-Heng Li*

Cite This: *Org. Lett.* 2024, 26, 7661−7666 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information



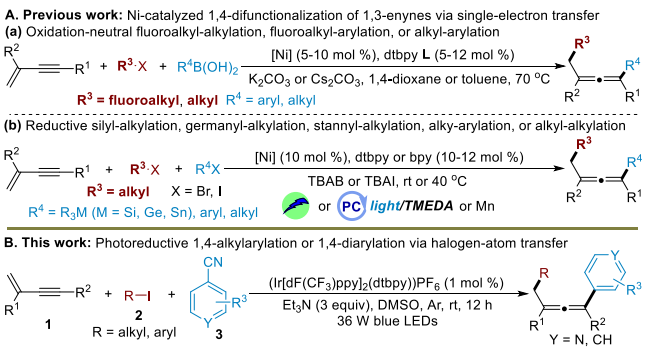
ABSTRACT: A photoreductive halogen-atom transfer (XAT) strategy for 1,4-dicarbofunctionalization of 1,3-enynes with organoiodides and cyanoarenes is disclosed, enabling access to functionalized allenes in a highly regio-, chemo-, and stereoselective manner. Upon the photoredox catalysis and the activation of Et_3N XAT agents, the mild conditions and high functional group tolerance of this protocol enable the formation of two C−C bonds, including a $C(sp^3)$−$C(sp^3)$ bond and a $C(sp^2)$−$C(sp^2)$ bond, in a single reaction step, and provides a general avenue to polysubstituted allenes and late-stage modification of bioactive compounds.

Downloaded via FUDAN UNIV on February 5, 2025 at 04:13:51 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

Allenes are valuable structural motifs frequently in natural products, bioactive molecules, and functional materials, as both end products and versatile building blocks.[1] For these reasons, much sustained attention has been paid to the development of efficient methodologies for building diverse functionlized allene frameworks.[2] Among them, the 1,3-enyne difunctionalization reaction has proven uniquely powerful to staightfoward access of allenes using readily accessible starting materials through simultaneously incorporating two functional groups into the 1,3-enyne systems with high levels of regio-, chemo-, and stereocontol.[2−4] Significant methods are 1,3-enyne alkylative 1,4-dicarbofunctionalization with alkyl halides via cross coupling, such as 1,4-dialkylation, 1,4-alkylarylation, and 1,4-diarylation, which allow incorporation of two carbofunctional groups to the 1,3-enynes leading to complex allenes (Scheme 1A).[3] These works are mainly focused on transition-metal (often nickel) catalysis for the redox-neutral 1,3-enyne 1,4-dicarbofunctionalization with alkyl halides and organometallics,[3a,d] reductive 1,3-enyne 1,4-dicarbofunctionalization with two electrophilic organohalides by cooperative metal reducing reagents, electroreductive catalysis, and/or photoreductive catalysis,[3e,h] thus giving rise to practical issues, such as the high cost of the catalytic system and the need for both prefunctionalized starting materials and limited terminating carbofunctional groups. Therefore, there is a need for developing mechanistically distinct strategies that allow incorporation of broad classes of readily accessible carbofunctional group feedstocks in 1,3-enyne systems without transition-metal catalysis to accomplish 1,3-enyne dicarbofunctionalization and assemble diverse functionalized allenes with high regio-, chemo-, and stereoselectivity.

Halogen atom transfer (XAT) is a fundamental elementary strategy, which engages the transformations of organohalides with nucleophilic XAT agents (such as traditional organotins, silicon reagents, trialkylborane-O_2 systems, and current alkylamines) into transposed carbon-centered radicals using thermochemical, photochemical, or electrochemical catalysis, in synthetic chemistry to fuel the mania for new reaction creation.[5] Particularly, recent achievements in photocatalytic methods employing alkylamines as XAT agents are effective for allowing C−X (X = halogen) functionalization of organohalides via XAT.[5e,l] To our knowledge, however, there are no reports documenting the successful use of XAT of organohalides to enable 1,4-difunctionalization of 1,3-enynes.

Scheme 1. 1,4-Difunctionalization of 1,3-Enynes



Received: July 27, 2024
Revised: August 21, 2024
Accepted: August 27, 2024
Published: August 28, 2024



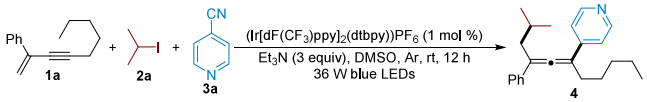
ACS Publications © 2024 American Chemical Society

https://doi.org/10.1021/acs.orglett.4c02774
Org. Lett. 2024, 26, 7661−7666

On this basis, we envisaged that using a tertiary amine as an electron transfer agent drives XAT of organohalides to form transposed carbon-centered radicals, which would implement regioselective addition across the alkene moiety, subsequent isomerization, and radical−radical coupling to realize 1,3-enyne 4-difunctionalization. Herein, we report a new photoreductive 1,4-alkylarylation of 1,3-enynes with alkyl iodides and cyanoarenes[6] via XAT for producing polysubstituted allenes in a highly regio-, stereo-, and chemoselective manner (Scheme 1B). Upon photoredox catalysis and activation of tertiary alkyl amine XAT agents, this method enables the formation of two C−C bonds, including a $C(sp^3)-C(sp^3)$ bond and a $C(sp^2)-C(sp^2)$ bond, in a single reaction step through single electron oxidation (SEO) to generate the amine radical cation, XAT, via radical addition across the C=C bonds, single electron reduction (SER), radical−radical coupling, and decyanation cascades, and provides a general avenue to polysubstituted allenes, utilizing readily accessible starting materials. Moreover, this strategy is applicable to 1,4-diarylation of 1,3-enynes with aryl iodides and cyanoarenes.

We began our optimization studies on the 1,4-alkylarylation of non-1-en-3-yn-2-ylbenzene **1a** with 2-iodopropane **2a** and isonicotinonitrile **3a** by visible light photoredox catalysis (Table 1). Using $(Ir[dF(CF_3)ppy]_2(dtbpy))PF_6$ (1 mol%) as

Table 1. Screening of the Optimal Reaction Conditions[a]



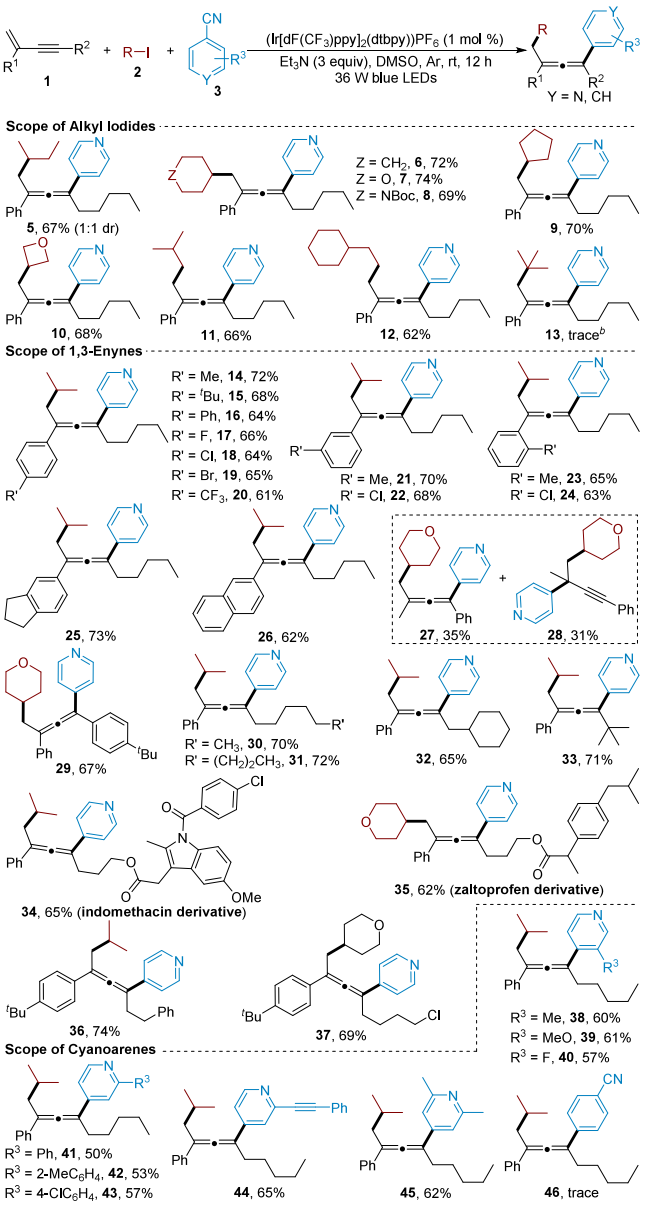
entry	variation from the standard conditions	yield (%)
1	none	71
2	without $(Ir[dF(CF_3)ppy]_2(dtbpy))PF_6$	0
3	using $[Ir(dtbbpy)(ppy)_2][PF_6]$	66
4	using $Ir(ppy)_3$	57
5	using $Ru(bpy)_3Cl_2$ or eosin Y	<5
6	using 4-CzIPN	61
7	without Et_3N	0
8	$^iPr_2NEt$ or TMEDA instead of Et_3N	55/45
9	Na_2CO_3 or Cs_2CO_3 instead of Et_3N	0
10	DMF, MeCN, or $PhCF_3$ instead of DMSO	35/51/18
11	without light (in dark)	0
12[b]	none	65

[a]Standard reaction conditions: **1a** (0.2 mmol), **2a** (0.3 mmol), **3a** (0.3 mmol), $(Ir[dF(CF_3)ppy]_2(dtbpy))PF_6$ (1 mol%), Et_3N (3 equiv), DMSO (2 mL), 36 W blue LEDs, argon, room temperature, and 12 h. [b]**1a** (1 mmol) and 12 h.

the photocatalyst under 36 W blue LED light in the presence of Et_3N and DMSO, the 1,4-alkylarylation of 1,3-enyne **1a** with iodide **2a** and cyanoarene **3a** operated smoothly, giving the desired product **4** in 71% yield (entry 1). After systematic screening of the reaction parameters, the results reveal that the parameters, including photocatalysts, Et_3N, and blue light, are essential for the reaction success, since omitting each resulted in no detectable product **4** (entries 2, 7, and 11). A series of photocatalysts, such as $[Ir(dtbbpy)(ppy)_2][PF_6]$, $Ir(ppy)_3$, $Ru(bpy)_3Cl_2$, eosin Y, and 4-CzIPN, were examined: they were less efficient than $(Ir[dF(CF_3)ppy]_2(dtbpy))PF_6$ (entries 3−6). Using other organic bases like $^iPr_2NEt$ and TMEDA instead of Et_3N executes the 1,3-enyne 1,4-alkylarylation smoothly, albeit with diminishing yields (entry 8), whereas handling an inorganic base, such as Na_2CO_3 and Cs_2CO_3, led to no reaction (entry 9). The results suggest that amines may serve as the electron agents (XAT agents). Other solvents, including DMF, MeCN, and $PhCF_3$, were found to be less reactive than DMSO (entry 10). Under the optimum reaction conditions, the reaction with a scale up to 1 mmol of **1a** operated efficiently and afforded **4** in 65% yield (entry 12).

With the optimum reaction conditions in hand, we set out to examine the scope of this photoreductive 1,3-enyne 1,4-alkylarylation protocol varying 1,3-enynes, organohalides, and cyanoarenes (Table 2 and Scheme 2). We found that secondary and primary alkyl iodides, especially those containing functional groups like cycloalkyl and heterocy-

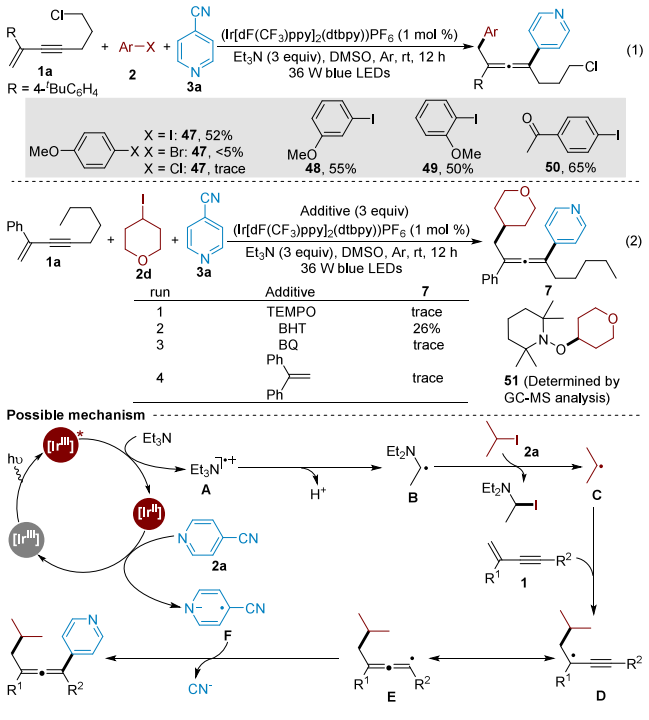
Table 2. Scope for the Alkylarylation of 1,3-Enynes (1) with Alkyl Iodides (2) and Cyanoarenes (3)[a]



[a]Reaction conditions: **1** (0.2 mmol), **2** (0.3 mmol), **3** (0.3 mmol), $(Ir[dF(CF_3)ppy]_2(dtbpy))PF_6$ (1 mol%), Et_3N (3 equiv), DMSO (2 mL), 36 W blue LEDs, argon, room temperature, and 12 h. [b]Using *t*-butyl iodide (0.3 mmol) or *t*-butyl bromide (0.3 mmol).

https://doi.org/10.1021/acs.orglett.4c02774
Org. Lett. 2024, 26, 7661−7666

Scheme 2. Aryl Halides, Control Experiments, and Possible Mechanism



cloalkyl, underwent the 1,3-enyne 1,4-alkylarylation reaction smoothly, affording **5**−**12** in good yields. However, tertiary alkyl halides such as *t*-butyl iodide and *t*-butyl bromide had no reactivity (**13**). Nonsymmetrical 2-iodobutane successfully delivered **5** in 67% yield and a 1:1 diastereoselectivity ratio (dr). Heterocycloalkyl iodides, such as 4-iodotetrahydro-2*H*-pyran, 4-iodopiperidine-1-carboxylate, and 3-iodooxetane, were efficiently converted to **7**, **8**, and **10**, respectively, in 68−74% yields. Gratifyingly, a wide range of 1,3-enynes were compatible with the optimum conditions (**14**−**37**). 1,3-Enynes with a monosubstituted aryl group, such as *p*-MeC_6H_4, *p*-^{t}BuC_6H_4, *p*-PhC_6H_4, *p*-FC_6H_4, *p*-ClC_6H_4, *p*-BrC_6H_4, *p*-$CF_3C_6H_4$, *m*-MeC_6H_4, and *o*-MeC_6H_4, at position 2 of the alkene moiety performed smoothly, giving **14**−**24** in 61−72% yields. Notably, the electron-donating aryl groups (**14**−**16**, **21**, **23**) are more reactive than the electron-withdrawing ones (**17**−**20**, **22**, **24**), and these products **17**−**19**, **22**, and **24** containing a halogen atom thus offer a potential handle for further modifications. 1,3-Enynes having an important substituent like 2,3-dihydro-1*H*-inden-5-yl and naphthalen-2-yl at position 2 were well tolerated (**25** and **26**). For (3-methylbut-3-en-1-yn-1-yl)-benzene bearing an aliphatic Me group at position 2, the protocol was successful for furnishing the desired alkene 1,4-alkylarylation product **27** in 35% yield along with 31% yield of the alkene 1,2-alkylarylation product **28**. The results support that 2-alkyl-1,3-enyne may be converted to both the tertiary benzyl radical and the less stable tertiary alkyl radical. 1,3-Enynes bearing an aryl or an alkyl group at the terminal alkyne are accommodated in the 1,3-enyne 1,4-alkylarylation protocol (**29**−**37**). Using 1,3-enyne containing a *p*-^{t}BuC_6H_4 group at the terminal alkyne was sufficient to assemble **29** in 67% yield. A number of alkyl substituents, including long-chain alkyl groups (*n*-hexyl, *n*-octyl), cyclohexylmethyl, *t*-butyl, propyl esters, 4-phenethyl, and *n*-chlorobutyl, at the terminal alkyne were well tolerated (**30**−**37**). This protocol is applicable to late-stage modification of bioactive molecules like indomethacin and zaltoprofen derivatives (**34**−**35**).[7]

Next, the versatility of this photoreductive 1,3-enyne 1,4-alkylarylation protocol was investigated by varying the cyanoarenes (**38**−**46**). A variety of cyanopyridines bearing a functionality, including Me, MeO, F, Ph, 2-MeC_6H_4, 4-ClC_6H_4, and phenylalkynyl, at the 2 or 3 position were converted to **38**−**44** smoothly in moderate to good yields. The results indicate that the electron-donating functionalities are more reactive than the electron-withdrawing ones. While isonicotinonitriles having an electron-donating group like Me or MeO furnished **38** and **39**, respectively, in 60−61% yields, the variant containing an electron-withdrawing F group delivered **40** in a slightly diminished yield. The important aryl and alkynyl functionalities can survive, assembling **41**−**44** in 50−65% yields. 2,6-DiMe-substituted isonicotinonitrile is a suitable substrate and afforded **45** in 62% yield.[8] Unfortunately, terephthalonitrile had no reactivity (**46**).

As shown in Scheme 2, we explored the feasibility of aryl halides for this 1,3-enyne 1,4-difunctionalization protocol (eq 1). We found that iodobenzene was viable for constructing **47** in 52% yield, but bromobenzene and chlorobenzene were not suitable substrates. Aryl iodides bearing a substituent, such as MeO and acetyl, at the *meta*, *ortho*, or *para* positions, could be converted to **48**−**50** in moderate to good yields. The control experiments with a radical scavenger such as TEMPO, BHT, BQ, and 1,1-diphenylalkene were examined. The reaction was completely inhibited when using TEMPO, BQ, or 1,1-diphenylalkene, providing the TEMPO-trapping product **51** (eq 2; Scheme 2). Using BHT also suppressed the reaction with a sharply diminishing yield. These experiments support the involvement of a radical process in this reaction.

The Stern−Volmer fluorescence quenching experiments of (Ir[dF(CF_3)ppy]$_2$(dtbpy))PF_6 with Et_3N, 2-iodopropane **2a**, and isonicotinonitrile **3a**, respectively, were investigated (Figures S1−S4; Supporting Information). Enhancing the concentration of Et_3N gradually quenched the fluorescence of (Ir[dF(CF_3)ppy]$_2$(dtbpy))PF_6, and increasing the concentration of **2a** or **3a** had no influence. These results imply that photocatalysts may preferentially react with Et_3N to form aminoalkyl radical cations. The light on−off experiments indicate that the 1,3-enyne difunctionalization reaction can not occur when the lamp is off during the process, suggesting that visible light is essential for the reaction success (Figure S5; Supporting Information). The cyclic voltammetry (CV) analysis of 2-iodopropane **2a**, isonicotinonitrile **3a**, or a combination of **2a**/Et_3N was conducted (Figures S6 and S7). We find that **2a** has a reversible reduction peak at $Ep_{1/2red}$ = −2.11 V vs standard calomel electrode (SCE), and **3a** showed a reversible reduction peak at $Ep_{1/2red}$ = −1.89 V vs SCE. Moreover, (Ir[dF(CF_3)ppy]$_2$(dtbpy))PF_6 has a reduction potential E_{red} = −1.37 V vs SCE.[8] These observations reveal that cathodic reduction of **3a** by the [Ir] photocatalyst has priority over that of **2a**. On the other hand, Et_3N exhibits an oxidative peak at $Ep_{1/2ox}$ = 0.99 V vs SCE, and 2-iodopropane **2a** appears as a reversible oxidative peak at $Ep_{1/2ox}$ = 2.28 V vs SCE. Notably, Et_3N combined with **2a** shows a lower oxidative potential at $Ep_{1/2ox}$ = 1.11 V vs SCE by comparison to 2-iodopropane **2a** with an oxidative peak at $Ep_{1/2ox}$ = 2.28 V vs SCE. The results suggest that Et_3N may be oxidized preferentially at the anode to readily generate the aminoalkyl radical cation, which would subsequently drive

https://doi.org/10.1021/acs.orglett.4c02774
Org. Lett. 2024, 26, 7661−7666

XAT with alkyl and aryl iodides to form the transposed carbon-centered radicals. We found that the quantum yield was determined to be $\Phi = 0.16$, implying that this reaction does not involve a radical chain process.

Based on the current mechanistic studies and literature precedents,[3−6] we propose the work mechanism for this photoreductive 1,3-enyne 1,4-dicarbonfunctionalization protocol (Scheme 2 (Possible mechanism)). Visible light irradiation of the $[Ir^{III}]$ species offers the reactive excited state $[Ir^{III}]^*$. Single electron oxidation (SEO) of Et_3N by the excited state $[Ir^{III}]^*$ gives rise to the formation of the amino radical cation intermediate **A** and the $[Ir^{II}]$ species, followed by XAT of iodide **2a** to afford the alkyl carbon-centered radical **B**.[5] Radical addition of intermediate **B** to the alkene produces the new alkyl radical intermediate **C**. During the process, single electron reduction (SER) of isonicotinonitrile **3a** by the $[Ir^{II}]$ species takes place to produce the isonicotinonitrile radical anion intermediate **D** and regenerate the active $[Ir^{III}]$ photocatalyst.[6] Finally, intermediate **C** undergoes radical−radical coupling with intermediate **D** and subsequent decyanation, accessing the desired allene.

In summary, we have showcased a novel visible-light-induced photoreductive 1,4-alkylarylation of 1,3-enynes with alkyl iodides and cyanoarenes for forging polysubstituted allenes in a highly regio-, chemo-, and stereoselective fashion. Upon the photoredox catalysis and the activation of tertiary alkyl amine XAT agents, this method enables the formation of two C−C bonds, including a $C(sp^3)-C(sp^3)$ bond and a $C(sp^2)-C(sp^2)$ bond, in a single reaction step through single electron oxidation to generate to the amine radical cation, XAT, via radical addition across the C═C bonds, single electron reduction, radical−radical coupling, and decyanation cascades, and provides a general avenue to polysubstituted allenes, utilizing the readily accessible starting materials.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c02774.

General experimental procedures, characterization details, NMR spectra, and X-ray crystallographic structure (PDF)

Accession Codes

CCDC 2326906 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

■ AUTHOR INFORMATION

Corresponding Authors

Yi-Lin Liu − *Hunan Engineering Laboratory for Preparation Technology of Polyvinyl Alcohol (PVA) Fiber Material, Institute of Organic Synthesis, Huaihua University, Huaihua 418000, China*; Email: liuyilinhn@126.com

Ming Hu − *State Key Laboratory Base of Eco-Chemical Engineering, College of Chemical Engineering, Qingdao University of Science and Technology, Qingdao 266042, China*; Email: huming0731@163.com

Jin-Heng Li − *Hunan Engineering Laboratory for Preparation Technology of Polyvinyl Alcohol (PVA) Fiber Material, Institute of Organic Synthesis, Huaihua University, Huaihua 418000, China; State Key Laboratory Base of Eco-Chemical Engineering, College of Chemical Engineering, Qingdao University of Science and Technology, Qingdao 266042, China; State Key Laboratory of Applied Organic Chemistry, Lanzhou University, Lanzhou 730000, China; School of Chemistry and Chemical Engineering, Henan Normal University, Xinxiang, Henan 475004, China*; orcid.org/0000-0001-7215-7152; Email: jhli@hnu.edu.cn

Authors

Liang-Feng Yang − *Hunan Engineering Laboratory for Preparation Technology of Polyvinyl Alcohol (PVA) Fiber Material, Institute of Organic Synthesis, Huaihua University, Huaihua 418000, China; State Key Laboratory Base of Eco-Chemical Engineering, College of Chemical Engineering, Qingdao University of Science and Technology, Qingdao 266042, China*

Liang Zeng − *State Key Laboratory Base of Eco-Chemical Engineering, College of Chemical Engineering, Qingdao University of Science and Technology, Qingdao 266042, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.4c02774

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank the National Natural Science Foundation of China (No 22271245), the Hunan Provincial Natural Science Foundation of China (No. 2024JJ7377), and the Open Research Fund of School of Chemistry and Chemical Engineering, Henan Normal University (No. 2021ZD01) for financial support.

■ REFERENCES

(1) (a) Zimmer, R.; Dinesh, C. U.; Nandanan, E.; Khan, F. A. Palladium-Catalyzed Reactions of Allenes. *Chem. Rev.* **2000**, *100*, 3067. (b) Modern Allene Chemistry; Krause, N., Hashmi, A. S. K., Eds.; Wiley-VCH: Weinheim, 2004. (c) Hoffmann-Roder, A.; Krause, N. Synthesis and Properties of Allenic Natural Products and Pharmaceuticals. *Angew. Chem., Int. Ed.* **2004**, *43*, 1196. (d) Ma, S. Some Typical Advances in the Synthetic Applications of Allenes. *Chem. Rev.* **2005**, *105*, 2829. (e) Rivera-Fuentes, P.; Diederich, F. Allenes in Molecular Materials. *Angew. Chem., Int. Ed.* **2012**, *51*, 2818. (f) Yu, S.; Ma, S. Allenes in Catalytic Asymmetric Synthesis and Natural Product Syntheses. *Angew. Chem., Int. Ed.* **2012**, *51*, 3074. (g) Ye, J.; Ma, S. Palladium-Catalyzed Cyclization Reactions of Allenes in the Presence of Unsaturated Carbon-Carbon Bonds. *Acc. Chem. Res.* **2014**, *47*, 989. (h) Liu, L.; Ward, R. M.; Schomaker, J. M. Mechanistic Aspects and Synthetic Applications of Radical Additions to Allenes. *Chem. Rev.* **2019**, *119*, 12422. (i) Alonso, J. M.; Almendros, P. Deciphering the Chameleonic Chemistry of Allenols: Breaking the Taboo of a Onetime Esoteric Functionality. *Chem. Rev.* **2021**, *121*, 4193.

(2) (a) Yu, S.; Ma, S. How Easy Are the Syntheses of Allenes? *Chem. Commun.* **2011**, *47*, 5384. (b) Lan, X.-W.; Wang, N.-X.; Xing, Y. Recent Advances in Radical Difunctionalization of Simple Alkenes. *Eur. J. Org. Chem.* **2017**, *2017*, 5821. (c) Chu, W.-D.; Zhang, Y.;

Wang, J. Recent Advances in Catalytic Asymmetric Synthesis of Allenes. *Catal. Sci. Technol.* **2017**, *7*, 4570. (d) Tu, H. Y.; Zhu, S.; Qing, F. L.; Chu, L. Recent Advances in Nickel-Catalyzed Three-Component Difunction- alization of Unactivated Alkenes. *Synthesis* **2020**, *52*, 1346. (e) Fu, L.; Greßies, S.; Chen, P.; Liu, G. Recent Advances and Perspectives in Transition Metal-Catalyzed 1,4-Functionaliza- tions of Unactivated 1,3-Enynes for the Synthesis of Allenes. *Chin. J. Chem.* **2020**, *38*, 91. (f) Li, Y.; Bao, H. Radical Transformations for Allene Synthesis. *Chem. Sci.* **2022**, *13*, 8491.

(3) (a) Zhang, K.-F.; Bian, K.-J.; Li, C.; Sheng, J.; Li, Y.; Wang, X.-S. Nickel-Catalyzed Carbofluoroalkylation of 1,3-Enynes to Access Structurally Diverse Fluoroalkylated Allenes. *Angew. Chem., Int. Ed.* **2019**, *58*, 5069. (b) Li, M.; Sun, G.-Q.; Liu, Y.-Y.; Li, S.-X.; Liu, H.-C.; Qiu, Y.-F.; Chen, D.-P.; Wang, X.-C.; Liang, Y.-M.; Quan, Z.-J. Nickel-Catalyzed Three-Component Tandem Radical Cyclization 1,5-Difunctionalization of 1,3-Enynes and Alkyl Bromide. *J. Org. Chem.* **2023**, *88*, 1403. (c) Shan, Q.-C.; Zhao, Y.; Wang, S.-T.; Liu, H.-F.; Duan, X.-H.; Guo, L.-N. Nickel-Catalyzed Modular Four-Component 1,4-Alkylcarbonylation of 1,3-Enynes to Tetra-Substituted CF_3-Allenyl Ketones. *ACS Catal.* **2024**, *14*, 2144. (d) Dong, X.-Y.; Zhan, T.-Y.; Jiang, S.-P.; Liu, X.-D.; Ye, L.; Li, Z.-L.; Gu, Q.-S.; Liu, X.-Y. Copper-Catalyzed Asymmetric Coupling of Allenyl Radicals with Terminal Alkynes to Access Tetrasubstituted Allenes. *Angew. Chem., Int. Ed.* **2021**, *60*, 2160. (e) Chen, H.; Zhu, C.; Yue, H.; Rueping, M. Group 14 Elements Hetero-Difunctionalizations via Nickel-Catalyzed Electro- reductive Cross-Coupling. *Angew. Chem., Int. Ed.* **2023**, *62*, e202306498. (f) Zhu, C.; Chen, H.; Yue, H.; Rueping, M. Electrochemical Chemo- and Regioselective Arylalkylation, Dialkylation and Hydro(deutero)alkylation of 1,3-Enynes. *Nat. Synth.* **2023**, *2*, 1068. (g) Wang, Q.; Chen, Y.; Peng, K.; Li, Y.; Cheng, L.; Deng, G.-J. Three-Component Cross-Electrophile 1,4-Alkylarylation of 1,3-Enynes by Merging Nickel and Photoredox Catalysis. *Org. Lett.* **2023**, *25*, 8889. (h) Jeon, J. H.; Kim, G. H.; Lee, H. S.; Kim, D. H.; Lee, S.; Choe, W.; Jung, B.; Rohde, J.-U.; Hong, S. Y. Chemo- and Regioselective Nickel-Catalyzed Reductive 1,4-Alkylarylation of 1,3-Enynes through an L_2NiAr Intermediate. *ACS Catal.* **2024**, *14*, 8996.

(4) For representative papers on the other 1,3-enyne difunctionalization reactions, see: (a) Wang, F.; Wang, D.; Zhou, Y.; Liang, L.; Lu, R.; Chen, P.; Lin, Z.; Liu, G. Divergent Synthesis of CF_3-Substituted Allenyl Nitriles by Ligand- Controlled Radical 1,2- and 1,4-Addition to 1,3-Enynes. *Angew. Chem., Int. Ed.* **2018**, *57*, 7140. (b) Zhu, X.; Deng, W.; Chiou, M.-F.; Ye, C.; Jian, W.; Zeng, Y.; Jiao, Y.; Ge, L.; Li, Y.; Zhang, X.; Bao, H. Copper-Catalyzed Radical 1,4-Difunctionalization of 1,3-Enynes with Alkyl Diacyl Peroxides and *N*-Fluorobenzenesulfonimide. *J. Am. Chem. Soc.* **2019**, *141*, 548. (c) Ye, C.; Li, Y.; Zhu, X.; Hu, S.; Yuan, D.; Bao, H. Copper-Catalyzed 1,4-Alkylarylation of 1,3-Enynes with Masked Alkyl Electrophiles. *Chem. Sci.* **2019**, *10*, 3632. (d) Zhang, Q.; Muhammad, M. T.; Chiou, M.-F.; Jiao, Y.; Bao, H.; Li, Y. 1,4-Fluoroamination of 1,3-Enynes en Route to Fluorinated Allenes. *Org. Lett.* **2020**, *22*, 5261. (e) Huang, J.; Jia, Y.; Li, X.; Duan, J.; Jiang, Z.-X.; Yang, Z. Halotrifluoromethylation of 1,3-Enynes: Access to Tetrasubstituted Allenes. *Org. Lett.* **2021**, *23*, 2314. (f) Song, Y.; Fu, C.; Ma, S. Copper-Catalyzed Syntheses of Multiple Functionalizatized Allenes via Three-Component Reaction of Enynes. *ACS Catal.* **2021**, *11*, 10007. (g) Chen, L.; Lin, C.; Zhang, S.; Zhang, X.; Zhang, J.; Xing, L.; Guo, Y.; Feng, J.; Gao, J.; Du, D. 1,4-Alkylcarbonylation of 1,3-Enynes to Access *Tetra*-Substituted Allenyl Ketones via an NHC-Catalyzed Radical Relay. *ACS Catal.* **2021**, *11*, 13363. (h) Cai, Y.; Chen, J.; Huang, Y. *N*-Heterocyclic Carbene-Catalyzed 1,4-Alkylacyl- ation of 1,3-Enynes. *Org. Lett.* **2021**, *23*, 9251. (i) Chen, Y.; Zhu, K.; Huang, Q.; Lu, Y. Regiodivergent Sulfonylarylation of 1,3-Enynes *via* Nickel/Photoredox Dual Catalysis. *Chem Sci.* **2021**, *12*, 13564. (j) Xu, T.; Wu, S.; Zhang, Q.-N.; Wu, Y.; Hu, M.; Li, J.-H. Dual Photoredox/Nickel-Catalyzed 1,4-Sulfonylarylation of 1,3-Enynes with Sulfinate Salts and Aryl Halides: Entry into Tetrasubstituted Allenes. *Org. Lett.* **2021**, *23*, 8455. (k) Li, C.; Hu, D.-D.; Jin, R.-X.; Wu, B.-B.; Wang, C.-Y.; Ke, Z.; Wang, X.-S. Selective 1,4-Arylsulfonation of 1,3-Enynes via Photoredox/Nickel Dual Catalysis. *Org Chem Front.* **2022**, *9*, 788. (l) Chen, Y.; Wang, J.; Lu, Y. Decarboxylative 1,4-Carbocyanation of 1,3-Enynes to Access Tetra-Substituted Allenes via Copper/Photoredox Dual Catalysis. *Chem Sci.* **2021**, *12*, 11316. (m) Sun, Q.; Zhang, X.-P.; Duan, X.; Qin, L.-Z.; Yuan, X.; Wu, M.-Y.; Liu, J.; Zhu, S.-S.; Qiu, J.-K.; Guo, K. Photoinduced Merging with Copper or Nickel Catalyzed 1,4 Cyanoalkylarylation of 1,3 Enynes to Access Multiple Functionalizatized Allenes in Batch and Continuous Flow. *Chin. J. Chem.* **2022**, *40*, 1537. (n) Zhang, F.-H.; Guo, X.; Zeng, X.; Wang, Z. Asymmetric 1,4-Functionalization of 1,3-Enynes via Dual Photoredox and Chromium Catalysis. *Nat Commun.* **2022**, *13*, 5036. (o) Wang, L.; Ma, R.; Sun, J.; Zheng, G.; Zhang, Q. NHC and Visible Light-Mediated Photoredox Co-Catalyzed 1,4-Sulfonylacylation of 1,3-Enynes for Tetrasubstituted Allenyl Ketones. *Chem Sci.* **2022**, *13*, 3169. (p) Liu, W.; Liu, C.; Wang, M.; Kong, W. Modular Synthesis of Multifunctionalized CF_3-Allenes through Selective Activation of Saturated Hydrocarbons. *ACS Catal.* **2022**, *12*, 10207. (q) Lv, Y.; Han, W.; Pu, W.; Xie, J.; Wang, A.; Zhang, M.; Wang, J.; Lai, J. Copper-Catalyzed Regioselective 1,4-Sulfonylcyanation of 1,3-Enynes with Sulfonyl Chlorides and TMSCN. *Org Chem Front.* **2022**, *9*, 3775. (r) Cheng, C.; Lv, G.-F.; Wu, S.; Li, Y.; Li, J.-H. Nickel Photocatalyzed Reductive 1,4-Dicarbofunction- alization of 1,3-Enynes with *N*-Methylamines and Organohalides Enabled by Site-Selective C(sp^3)-H Functionalization. *Org. Lett.* **2023**, *25*, 4236. (s) Pu, Y.; Ding, S.; Zhao, H.; Xue, Q.; Zhang, H.; Xie, X.; Shang, Y.; Wang, J. Three-Component Synthesis of Multiple Functionalized Allenes via Copper/Photoredox Dual Catalyzed 1,4-Alkylcyanation of 1,3-Enynes. *Org. Lett.* **2024**, *26*, 1834. (t) Yang, L.; Zeng, L.; Zhu, Y.; Wang, Q.; Li, J. Copper-Catalyzed Photoredox 1,4-Amidocyanation of 1,3-Enynes with *N*-Amidopyridin-1-ium Salts and TMSCN: Facile Access to α-Amido Allenyl Nitriles. *Chin. Chem. Lett.* **2024**, *35*, 109685. (u) Li, Q.; Wang, Z.-L.; Xu, Y.-H. Copper-Catalyzed 1,4-Silylcyanation of 1,3-Enynes: A Silyl Radical-Initiated Approach for Synthesis of Difunctionalized Allenes. *Chin. Chem. Lett.* **2024**, *35*, 109991. (v) Xia, J.; Ma, R.; Wang, L.; Sun, J.; Zheng, G.; Zhang, Q. NHC and Photoredox Catalysis Dual-Catalyzed 1,4-Mono-/di-fluoro- methylative Acylation of 1,3-Enynes. *Org. Chem. Front.* **2024**, *11*, 3089.

(5) For selected reviews and representative papers, see: (a) Chatgilialoglu, C.; Ferreri, C.; Landais, Y.; Timokhin, V. I. Thirty Years of $(TMS)_3SiH$: A Milestone in Radical-Based Synthetic Chemistry. *Chem. Rev.* **2018**, *118*, 6516. (b) Yao, J.; Yan, J. Interfacial Polarization Triggered by Single Atoms Boosts N_2 Electroreduction. *Chem* **2020**, *6*, 808. (c) Juliá, F.; Constantin, T.; Leonori, D. Applications of Halogen-Atom Transfer (XAT) for the Generation of Carbon Radicals in Synthetic Photochemistry and Photocatalysis. *Chem. Rev.* **2022**, *122*, 2292. (d) Piedra, H. F.; Valdés, C.; Plaza, M. Shining Light on Halogen-Bonding Complexes: A Catalyst-Free Activation Mode of Carbon-Halogen Bonds for the Generation of Carbon-Centered Radical. *Chem. Sci.* **2023**, *14*, 5545. (e) Sachidanandan, K.; Niu, B.; Laulhé, S. An Overview of α Aminoalkyl Radical Mediated Halogen Atom Transfer. *ChemCatChem* **2023**, *15*, e202300860. (f) Constantin, T.; Zanini, M.; Regni, A.; Sheikh, N. S.; Juliá, F.; Leonori, D. Aminoalkyl Radicals as Halogen-Atom Transfer Agents for Activation of Alkyl and Aryl Halides. *Science* **2020**, *367*, 1021. (g) Tian, X.; Kaur, J.; Yakubov, S.; Barham, J. P. α-Amino Radical Halogen Atom Transfer Agents for Metallaphotoredox-Catalyzed Cross-Electrophile Couplings of Distinct Organic Halides. *ChemSusChem* **2022**, *15*, e202200906. (h) Zeng, L.; Qin, J.-H.; Lv, G.-F.; Hu, M.; Sun, Q.; Ouyang, X.-H.; He, D.-L.; Li, J.-H. Electrophotocatalytic Reductive 1,2-Diarylation of Alkenes with Aryl Halides and Cyanoaromatics. *Chin. J. Chem.* **2023**, *41*, 1921. (i) Maust, M. C.; Blakey, S. B. Photoredox-Driven Three-Component Coupling of Aryl Halides, Olefins, and O_2. *ACS Catal.* **2024**, *14*, 2582. (j) Zeng, L.; Xu, C.-H.; Zou, X.-Y.; Sun, Q.; Hu, M.; Ouyang, X.-H.; He, D.-L.; Li, J.-H. Iodoarene-Directed Photoredox β-C(sp^3)-H Arylation of 1-(*o*-Iodoaryl)alkan-1-ones with Cyanoarenes via Halogen Atom Transfer and Hydrogen Atom Transfer. *Chem. Sci.* **2024**, *15*, 6522. (k) Zeng, L.; Ren, H.-Z.; Lv, G.-F.; Ouyang, X.-H.; He, D.-L.; Li, J.-H. Electroreductive Remote Benzylic C(sp^3)-H Arylation of Aliphatic Ethers Using Cyanoarenes for the Synthesis of α-(Hetero)aryl Ethers.

https://doi.org/10.1021/acs.orglett.4c02774
Org. Lett. 2024, 26, 7661−7666

Org. Lett. **2024**, *26* (12), 2440. (l) Sephton, T.; Large, J. M.; Natrajan, L. S.; Butterworth, S.; Greaney, M. F. XAT-Catalysis for Intramolecular Biaryl Synthesis. *Angew. Chem., Int. Ed.* **2024**, *63*, e202407979.

(6) Tong, S.; Li, K.; Ouyang, X.; Song, R.; Li, J. Recent Advances in the Radical-mediated Decyanative Alkylation of Cyano(hetero)arene. *Green Synth. Catal.* **2021**, *2*, 145.

(7) (a) Surwase, S. A.; Boetker, J. P.; Saville, D.; Boyd, B. J.; Gordon, K. C.; Peltonen, L.; Strachan, C. J. Indomethacin: New Polymorphs of an Old Drug. *Mol. Pharmaceutics* **2013**, *10*, 4472. (b) Hirate, K.; Uchida, A.; Ogawa, Y.; Arai, T.; Yoda, K. Zaltoprofen, A Non-steroidal Anti-Inflammatory Drug, Inhibits Bradykinin-Induced Pain Responses Without Blocking Bradykinin Receptors. *Neurosci. Res.* **2006**, *54*, 288.

(8) CCDC 2326906 (**45**) contains the supplementary crystallographic data for this study. These data can be obtained free of charge from the Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

https://doi.org/10.1021/acs.orglett.4c02774
Org. Lett. 2024, 26, 7661−7666